

PARTIE 8 —

vérification et screenshot

1. Les sites d'un même client communiquent entre eux.

Ping vrf CAN 172.16.1.1 source f0/0 marche

```
PE2#PIN
PE2#PING vr
PE2#PING vrf CAN 172.16.1.1 SOU
PE2#PING vrf CAN 172.16.1.1 SOUrcE F0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
Packet sent with a source address of 30.0.0.2
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/48/60 ms
PE2#
```

2. Les clients de sites differents ne peuvent pas communiquer entre eux.

Ping vrf CAN 172.17.1.1 source f0/0 ne marche pas

```
PE2#
PE2#PING vrf CAN 172.17.1.1 SOUrcE F0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.1.1, timeout is 2 seconds:
Packet sent with a source address of 30.0.0.2
.....
Success rate is 0 percent (0/5)
PE2#
```

Show ip vrf

```
PE2#SH IP VR
PE2#SH IP VRf
  Name                Default RD      Interfaces
  CAN                  100:2          Fa0/0
  WAL                  200:2          Fa0/1
PE2#
PE2#
```

4.le trafic passe sur le CORE en utilisant les label LDP

Show mpls forwarding table

```
PE2#SH MPLs fo
PE2#SH MPLs forwarding-table
Local      Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id Switched      interface
200        500        1.1.1.1/32  0            Se1/0      point2point
201        502        4.4.4.4/32  0            Se1/0      point2point
202        503        5.5.5.5/32  0            Se1/0      point2point
203        Pop Label  6.6.6.6/32  0            Se1/0      point2point
204        504        7.7.7.7/32  0            Se1/0      point2point
205        505        60.0.0.0/30 0            Se1/0      point2point
206        506        70.0.0.0/30 0            Se1/0      point2point
207        507        80.0.0.0/30 0            Se1/0      point2point
208        Pop Label  90.0.0.0/30 0            Se1/0      point2point
209        Pop Label  100.0.0.0/30 0           Se1/0      point2point
210        No Label  30.0.0.0/30[V] 1710        aggregate/CAN
211        No Label  40.0.0.0/30[V] 570         aggregate/WAL
212        No Label  172.16.2.1/32[V] 3624        Fa0/0      30.0.0.1
213        No Label  172.17.2.0/24[V] 0           Fa0/1      40.0.0.1
PE2#
PE2#
```

Show mpls ldp neighbor

```
PE2#SH MPLs ld
PE2#SH MPLs ldp ne
PE2#SH MPLs ldp neighbor
  Peer LDP Ident: 6.6.6.6:0; Local LDP Ident 2.2.2.2:0
    TCP connection: 6.6.6.6.49081 - 2.2.2.2.646
    State: Oper; Msgs sent/rcvd: 225/226; Downstream
    Up time: 03:04:42
    LDP discovery sources:
      Serial1/0, Src IP addr: 50.0.0.1
    Addresses bound to peer LDP Ident:
      50.0.0.1      90.0.0.1      100.0.0.2      6.6.6.6
PE2#
```

5.la table de routage VRF est complete

Show ip route vrf WAL

Show ip route vrf CAN

```
PE2#show o
PE2#show ip route vr
PE2#show ip route vrf WAL

Routing Table: WAL
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override

Gateway of last resort is not set

      20.0.0.0/30 is subnetted, 1 subnets
B       20.0.0.0 [200/0] via 1.1.1.1, 00:41:29
      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/30 is directly connected, FastEthernet0/1
L       40.0.0.2/32 is directly connected, FastEthernet0/1
      172.17.0.0/24 is subnetted, 2 subnets
B       172.17.1.0 [200/156160] via 1.1.1.1, 00:41:29
D       172.17.2.0 [90/156160] via 40.0.0.1, 01:32:23, FastEthernet0/1
PE2#
PE2#
PE2#show ip route vrf CAN

Routing Table: CAN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/30 is subnetted, 1 subnets
B       10.0.0.0 [200/0] via 1.1.1.1, 02:00:49
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/30 is directly connected, FastEthernet0/0
L       30.0.0.2/32 is directly connected, FastEthernet0/0
      172.16.0.0/32 is subnetted, 2 subnets
B       172.16.1.1 [200/2] via 1.1.1.1, 02:00:49
O       172.16.2.1 [110/2] via 30.0.0.1, 01:34:08, FastEthernet0/0
PE2#
```

6.la table lib montre les réseaux des clients

Show ip bgp vpnv4 all

```
PE1#
PE1#SH IP Bgp Vpnv4 All
BGP table version is 15, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf CAN)
*> 10.0.0.0/30          0.0.0.0              0           32768 ?
*>i 30.0.0.0/30          2.2.2.2              0          100      0 ?
*> 172.16.1.1/32        10.0.0.1             2          32768 ?
*>i 172.16.2.1/32        2.2.2.2              2          100      0 ?
Route Distinguisher: 100:2
*>i 30.0.0.0/30          2.2.2.2              0          100      0 ?
*>i 172.16.2.1/32        2.2.2.2              2          100      0 ?
Route Distinguisher: 200:1 (default for vrf WAL)
*> 20.0.0.0/30          0.0.0.0              0           32768 ?
*>i 40.0.0.0/30          2.2.2.2              0          100      0 ?
*> 172.17.1.0/24        20.0.0.1            156160     32768 ?
*>i 172.17.2.0/24        2.2.2.2             156160     100      0 ?
Route Distinguisher: 200:2
*>i 40.0.0.0/30          2.2.2.2              0          100      0 ?
*>i 172.17.2.0/24        2.2.2.2             156160     100      0 ?
PE1#
PE1#

PE2#SHOW IP Bgp VP
PE2#SHOW IP Bgp Vpnv4 ALL
BGP table version is 13, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 100:1
*>i 10.0.0.0/30          1.1.1.1              0          100      0 ?
*>i 172.16.1.1/32        1.1.1.1              2          100      0 ?
Route Distinguisher: 100:2 (default for vrf CAN)
*>i 10.0.0.0/30          1.1.1.1              0          100      0 ?
*> 30.0.0.0/30          0.0.0.0              0           32768 ?
*>i 172.16.1.1/32        1.1.1.1              2          100      0 ?
*> 172.16.2.1/32        30.0.0.1             2          32768 ?
Route Distinguisher: 200:1
*>i 20.0.0.0/30          1.1.1.1              0          100      0 ?
*>i 172.17.1.0/24        1.1.1.1            156160     100      0 ?
Route Distinguisher: 200:2 (default for vrf WAL)
*>i 20.0.0.0/30          1.1.1.1              0          100      0 ?
*> 40.0.0.0/30          0.0.0.0              0           32768 ?
*>i 172.17.1.0/24        1.1.1.1            156160     100      0 ?
*> 172.17.2.0/24        40.0.0.1            156160     32768 ?
PE2#
PE2#
PE2#
PE2#
PE2#
```

7.le trafic qui sort du loopback du client passe par le vrf et aussi il a un label du mpls sur CE-A1

traceroute 172.16.2.1 source lo0

CE-A1#

CE-A1#

CE-A1#traceroute 172.16.2.1 source loopback 0

Type escape sequence to abort.

Tracing the route to 172.16.2.1

VRF info: (vrf in name/id, vrf out name/id)

1 10.0.0.2 12 msec 8 msec 12 msec

2 60.0.0.1 [MPLS: Labels 301/212 Exp 0] 64 msec 52 msec 52 msec

3 100.0.0.2 [MPLS: Labels 501/212 Exp 0] 52 msec 64 msec 52 msec

4 30.0.0.2 [MPLS: Label 212 Exp 0] 40 msec 44 msec 40 msec

5 30.0.0.1 52 msec 52 msec 52 msec

CE-A1#